Arab Academy for Science, Technology & Maritime Transport

Customer Churn in Digital Banking: An AI-Enabled Case Study of Crédit Agricole Egypt

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Abstract

Based on my practical experience in banking, customer churn is usually a gradual process rather than a sudden decision. This Report develops a full case study on customer churn in digital banking with specific application to Crédit Agricole Egypt. The study is structured to match the required report sequence in the Smart Village Campus MBA brief, moving from industry context and problem framing through data analytics, AI solution design, digital transformation, ERP integration, business value, risks, and future recommendations. The analysis is grounded in a realistic simulated dataset of 20,000 customers distributed across seven branches, with demographic, transactional, lifecycle, and risk attributes. The baseline dataset records 3,165 churned customers and an overall churn rate of 15.82%, which is sufficiently high to justify a dedicated retentionn programme. Statistical profiling shows that churn is concentrated among low-balance and individual customers, while average balances and average customer lifetime value are materially lower among churned customers than active customers.

The report combines descriptive, diagnostic, predictive, and prescriptive analytics. A Power BI dashboard is used to visualize the current-state churn profile, monthly trend, branch distribution,

and churn reasons. A logistic-regression-based supervised learning design is then proposed to move the bank from lagging, report-based monitoring to proactive risk scoring. On the simulated dataset, the model achieves high discrimination, and the strongest practical signals are days since last transaction, engineered risk score, low-balance profile, and weak digital engagement. Because the dataset already contains a risk score and behaviourally meaningful variables, the model is treated as a decision-support engine rather than a fully autonomous decision maker.

The paper concludes that Crédit Agricole Egypt can materially improve retention if it integrates CRM, core banking, and digital behaviour data into an AI-enabled workflow. A target-state scenario based on a 25% reduction in churn is used to demonstrate the potential uplift in retention KPIs, branch actions, and resource allocation. The report also addresses governance, privacy, ISO 27001-aligned controls, and the limitations of simulated data. Overall, the case shows that churn management in digital banking is not only a modelling problem but an enterprise transformation problem requiring data quality, process redesign, customer experience improvement, and continuous monitoring.

Keywords: customer churn, digital banking, Crédit Agricole Egypt, predictive analytics, logistic regression, Power BI, ERP integration, digital transformation, ISO 27001.

Project Objective and Case Scope

The main objective of this case study is to critically analyse how AI can decrease customer churn percentage in a digital banking environment. The scope is deliberately aligned to the MBA case-study requirements: it starts by locating the problem within the Egyptian banking industry, then explains why traditional systems are insufficient, demonstrates how data analytics and AI can be applied, evaluates process and ERP implications, and ends with business impact, risk controls, and future recommendations. The report is written as a unified paper rather than as disjointed bullet answers, but each requirement from Topic 3 to Topic 10 is explicitly covered through section numbering and analytical discussion.

3. Industry, Business Context, and Problem Framing

3.1 Industry and Business Context

The banking sector in Egypt is operating in a period of structural transition. Digital channels are no longer peripheral delivery mechanisms; they are becoming the primary interface through which

many customers evaluate service quality, convenience, and responsiveness. This change matters because customer loyalty in banking, historically built on branch familiarity and relationship with the front liners employees' continuity, is now increasingly shaped by digital functionality, speed of issue resolution, quality of mobile applications, and the ease of switching to a competitor. In practical terms, banks compete not only with other licensed banks but also with fintech providers and digital-first financial platforms that set new standards for convenience and personalisation.

Within this environment, Crédit Agricole Egypt can be characterised as a universal bank with retail, payroll, corporate, and SME-facing activity, operating through a branch network and digital channels. For a bank of this type, churn is not only a marketing metric. It is the key to its sustainability and a business health signal that affects deposit stability, fee income, cross-sell potential, and the economics of customer acquisition. When customers become inactive, stop using their primary account, transfer salary relationships, or close accounts entirely, the bank loses more than one balance line; it loses future revenue opportunities and weakens the depth of the relationship.

The realistic dataset used in this case reflects this context. It contains many variables like {summary_stats['total_customers']:,} customer records with status classes of Active, Inactive, and Churned. The overall churn rate of {summary_stats['churn_rate']:.2f}% and the additional inactive share of {summary_stats['inactive_rate']:.2f}% indicate that the bank's challenge is not confined to completed churn. A broader disengagement problem exists upstream of closure and should therefore be managed as a pipeline of risk rather than a single end-state event. This is precisely why the case is suited to an AI-in-business analysis: the value of AI lies in identifying customers while intervention is still possible.

3.2 Overview of the Selected Industry and Organization Type

The selected industry is digital banking within the wider banking and financial services sector. Unlike subscription industries where churn is often visible through formal cancellation, banking churn is multi-form. A customer may close an account, leave a salary relationship, reduce transactional intensity to near-zero, or continue holding an account while effectively moving primary banking activity elsewhere, or not qualified staff. This makes churn harder to observe and more expensive to ignore. A bank may believe a customer is retained because the account remains open, while in reality the relationship value has already decayed.

Crédit Agricole Egypt, as the organization type in this case, sits at the intersection of traditional banking infrastructure and modern digital service expectations. It relies on core banking systems for account and transaction records [flexcube] , CRM for customer information and service actions, and digital channels for ongoing engagement. As a result, the organization is especially exposed to the consequences of fragmented data. If digital usage, complaint history, balance movement, and account status are viewed separately, managers see isolated events. If those elements are joined analytically, managers can recognise a churn pattern. In that sense, the organization type is ideal for a case study on AI because it has sufficient structured data to support predictive modelling, yet it also faces the practical enterprise challenge of integrating insights into day-to-day operations.

This case therefore treats Crédit Agricole Egypt not merely as a brand example but as an operationally plausible banking setting in which an AI use case can be tested against business reality: branch variation, customer-type variation, risk segmentation, and retention constraints.

3.3 Key Business Challenges Faced Before AI Adoption

Before AI adoption, the principal challenge was the bank's dependence on lagging indicators. Traditional churn tracking answers the question, "Who left?" rather than the more valuable question, "Who is likely to leave next?" In the dataset and dashboard used for this project, churned customers already display recognizable traits: lower balances, higher risk scores, longer periods since last transaction, and concentration in certain customer segments. The fact that these patterns are visible after analysis implies that they could have been identified earlier if the right analytical workflow had existed.

A second challenge was fragmentation. In practice, customer information is distributed across multiple systems such as CRM, core banking, digital usage reporting, and complaint management. This means branch teams and decision makers do not naturally receive a single, integrated churn view. Instead, they depend on manual extraction and reconciliation. Manual processes slow down action, introduce inconsistency, and make it difficult to scale personalized intervention.

A third challenge concerns prioritisation. Without a churn model or structured risk framework, retention efforts tend to be generic. High-value customers and low-value customers may receive the same communication, or worse, no communication at all until the relationship has materially

deteriorated. The current data indicates that average balance for churned customers is only EGP {summary_stats['avg_balance_churned']:,.2f}, compared with EGP {summary_stats['avg_balance_active']:,.2f} for active customers. That gap is analytically useful: it signals that declining relationship depth is visible in advance. Yet in a pre-AI environment, such signals are rarely operationalized consistently.

3.4 Market Conditions and Competitive Environment

The market conditions surrounding customer churn in Egypt can be summarised by three forces: ease of switching, digital comparison, and service expectations. First, customers can increasingly transfer transaction behaviour away from a bank without immediate formal closure. Second, digital channels make service comparisons more visible and it’s the main factoers of competitio. Third, competitors can attract disaffected customers with better onboarding, simpler app experiences, fee reductions, and targeted payroll or salary propositions.

The current dataset reinforces this market logic. Individual customers make up the largest segment and also carry the highest churn rate. In the simulated data, the churn rate for Individual customers reaches 19.31%, compared with 8.23% for Payroll customers and 2.99% for corporate customers. This give the management something strategically important: relationship embeddedness matters. Payroll and corporate relationships are harder to dislodge because they are tied to recurring transactional behaviour, organisational links, or broader banking arrangements. Individual customers, by contrast, are more exposed to service dissatisfaction, poor digital experience, and competitor offers.

Branch-level variation also matters. The highest churn rates in the sample appear in Nasr City, Alexandria, Maadi, and 6th October. That does not automatically mean those branches are underperforming; branch context may reflect customer mix, salary competition, local service issues, or the digital adoption profile of customers served there. However, it does mean the bank should avoid treating churn as a uniform enterprise problem. Market conditions are filtered through branch realities, and any AI-enabled solution must allow local prioritisation rather than one national retention message for everyone.

3.5 Problem Statement

The business problem in this case can be stated clearly as follows: Crédit Agricole Egypt needs a way to identify customers who are at high risk of churn early enough to intervene effectively, while allocating retention resources based on customer risk and value rather than intuition or static reporting. The bank's traditional reporting environment is able to describe churn after it occurs but is insufficient for preventing it.

In other words, the core issue is not a lack of data, but a lack of integrated predictive use of data. The bank already has access to behavioral signals such as transaction recency, balance pattern, status transitions, customer type, branch, and lifetime value proxies. What is missing is the analytical and operational framework that converts those signals into decisions. This is why the problem naturally extends beyond data science into business process redesign, CRM execution, ERP integration, and governance.

3.6 Why Traditional Systems or Decision-Making Methods Were Insufficient

Traditional methods are insufficient for three reasons. First, they are primarily descriptive. A static dashboard or periodic report can tell managers that churn was high in January or July, but it cannot prioritise which active customers need outreach this week. Second, manual analysis does not scale. If branch managers or relationship staff must inspect lists manually, the process becomes selective, inconsistent, and too slow for thousands of customer records. Third, traditional approaches struggle with interaction effects. A customer who is young, individual, low-balance, digitally disengaged, and showing long transaction inactivity is much riskier than a customer who exhibits only one of those traits. Predictive models are useful precisely because they combine multiple features into a probability estimate.

The current dashboard image supplied for the case confirms both the value and the limitation of traditional reporting. It visualises churn rate, number of churned customers, customer volume, monthly variation, and churn reasons. That is analytically useful. However, on its own it remains a rear-view mirror. It does not identify who should be contacted, what offer should be triggered, or which branch portfolio should be escalated first. AI addresses that gap by turning descriptive patterns into forward-looking decision support.

3.7 Operational, Strategic, and Financial Gaps

Operationally, the bank faces a gap in response speed. Churn risk signals emerge before account closure, but there is no systematic rule to capture them and trigger action. Strategically, the gap lies in customer prioritisation. Retention decisions need to distinguish between customers whose value is low and recoverable and customers whose value is high and whose departure would materially weaken portfolio economics. Financially, the gap appears in foregone lifetime value and avoidable acquisition cost. Industry literature consistently argues that retaining existing customers is more cost-effective than replacing them, and that principle is especially relevant here because average CLV in the dataset is markedly lower among churned customers than active customers.

In this case, the average CLV of churned customers is approximately EGP {summary_stats['avg_clv_churned']:,.2f}, whereas active customers average approximately EGP {summary_stats['avg_clv_active']:,.2f}. The difference does not imply that all high-value customers are safe; rather, it implies that attrition is already eroding relationship value before a customer fully exits. This gives the bank a financial reason to act earlier. The gap therefore is not simply “too much churn.” The deeper gap is “not enough analytically prioritised intervention before value leakage becomes irreversible.”

3.8 Data Analytics Component

The data analytics component is mandatory in the case brief, and in this report it is treated as the empirical foundation of every later section. The dataset is predominantly structured and historical. Structured data appears in defined columns such as balance, branch, customer type, churn reason, CLV, status, and risk score. Historical data appears through last transaction date and the monthly summary that records the bank’s customer base, active customers, inactive customers, churned customers, and monthly churn rate over a twelve-month horizon. Unstructured data is not directly modelled in the current file, but the case recognises that complaint text, call-center notes, and sentiment signals would strengthen future versions of the solution.

The primary data sources represented in the case are conceptually drawn from core banking, CRM, and digital banking channels. Core banking provides balance, account status, and transaction dates; CRM contributes customer identity, segmentation, and potential service interactions; digital systems contribute behavioural engagement signals, even if only indirectly reflected in the current risk score and churn labels. These sources are sufficient to support a first-generation churn programme,

particularly because the data is already customer-level rather than branch-aggregate only.

Table 1 summarises the baseline dataset profile used in the study.

Table 1. Baseline dataset profile

Metric	Value
Total customers	20,000
Active customers	11,895
Inactive customers	4,940
Churned customers	3,165
Overall churn rate	15.82%
Average balance (all customers)	EGP 75,423.88
Average balance (active)	EGP 94,900.05
Average balance (churned)	EGP 35,056.71
Average risk score (active)	27.46
Average risk score (churned)	95.29

Several early conclusions arise from these baseline statistics. First, churn and inactivity together account for a large share of the portfolio, which means retention should not be defined narrowly as account closure prevention. Second, the balance and risk-score gap between active and churned customers supports the intuition that churn is preceded by weaker relationship depth and visible deterioration signals. Third, because the dataset contains explanatory variables that are operationally plausible, it is suitable for predictive modelling and not merely for historical reporting.

4. AI Solution Overview

4.1 Description of the AI Technology Used

The primary AI technology used in this case is supervised machine learning for binary classification, with logistic regression chosen as the baseline model. This choice is appropriate for an MBA case study for two reasons. First, logistic regression is interpretable; managers can understand the direction and relative contribution of features more easily than with more opaque

methods. Second, the business requirement is to estimate the probability that a customer belongs to the churn class, not to generate an abstract score with no decision context.

The classification framework is straightforward: the target variable is whether a customer churned, and the explanatory variables include age, balance, customer type, balance segment, branch, CLV, risk score, and days since last transaction. The model therefore learns patterns already embedded in customer behaviour. In a production environment, the bank would retrain this model periodically and compare it with more complex algorithms such as random forests, gradient boosting, or XGBoost. For the present case, logistic regression is sufficient to demonstrate how AI can move the institution from broad trend awareness to customer-level prediction.

4.2 Business Functions Impacted

The AI solution affects multiple business functions rather than only one analytics team. In marketing and CRM, churn prediction changes campaign design. Instead of launching broad campaigns across entire segments, the bank can target customers with elevated risk scores and tailor communication by branch, customer type, and likely churn reason. In branch and relationship management, the output supports portfolio prioritisation. Staff can focus on customers whose probability of churn is high enough to justify outreach and whose value justifies retention cost.

In finance, the model affects revenue forecasting and customer lifetime value preservation. Because churn erodes future balances, fee income, and cross-sell opportunity, the ability to reduce churn improves forecast quality and customer economics. In operations, AI reduces the manual effort of list preparation and exception checking. Finally, in digital banking and customer experience functions, churn prediction highlights where product and service quality need improvement. For example, if app issues and weak engagement coincide with high churn, the problem is not only whom to contact; it is also what to fix in the digital experience.

4.3 Explanation of How AI Addresses the Defined Problem

AI addresses the defined problem by converting raw customer signals into a ranked intervention list. Without AI, managers may know that churn is high but still be unable to identify which specific customers are entering danger. With AI, the bank can estimate individual churn probability and move customers into actionable buckets such as High Risk, Medium Risk, and Low Risk. This supports prioritised intervention, branch accountability, and retention budget discipline.

The solution is especially valuable because the churn process in banking is gradual. Customers often become less active, reduce balances, disengage from digital channels, complain, or respond to competitive offers before formally exiting. AI is effective because it recognises these combinations earlier than manual review. It does not replace business judgment. Rather, it gives business teams a disciplined way to allocate attention. The practical output is not simply a percentage probability but a workflow: identify, prioritise, intervene, and monitor response.

4.4 Explain MLM (Supervised, Unsupervised, and Semi-Supervised)

In this case study , supervised learning is the primary and current method because the dataset includes labelled outcomes. The model can clearly see which customers churned and which did not, making it suitable for classification. Unsupervised learning remains highly relevant, even if it is not the main model in the current design. It could be used to cluster customers into behavioural groups—for example, digitally engaged payroll customers, low-balance dormant individual customers, or branch-specific at-risk cohorts. Such segmentation would enrich campaign strategy and make retention offers more context specific.

Semi-supervised learning is a potential future enhancement. In a real bank, not every at-risk customer has a fully confirmed churn outcome immediately, and some data elements may be incomplete. Semi-supervised techniques could use a small labelled set alongside a larger unlabelled set to improve learning efficiency. For the current report, however, the important conclusion is that supervised learning is the correct choice for the case study because the business objective is prediction of a known outcome, and the available data supports that objective directly.

4.5 Explain RPA in the Case Study

Robotic Process Automation is not the predictive engine in this case; it is the execution layer that turns predictive insight into operational action. Once the model identifies high-risk customers, RPA can automate repetitive follow-up steps. These may include generating branch worklists, pushing records into CRM queues, sending customer notifications, assigning outreach tasks to relationship managers, or triggering predefined offers such as fee waivers, service calls, or digital education prompts.

RPA matters because one of the classic failure points in analytics projects is the gap between insight

and action. A bank may build a strong model but still fail to reduce churn if no operational process consumes the output consistently. By embedding RPA into the solution, the bank ensures that a high-risk score does not remain in a dashboard; it initiates a response. In other words, AI predicts and RPA operationalizes.

4.6 SDLC Methodology for the Case

The Software Development Life Cycle provides the project discipline needed to convert a case-study model into an enterprise solution. In the planning phase, the bank defines churn reduction as the business objective and clarifies the intervention use case. In the analysis phase, data fields, business rules, branch inputs, and reporting requirements are reviewed. In the design phase, feature logic, model structure, dashboard requirements, and action workflows are specified. Development includes data preparation, feature engineering, model training, and dashboard assembly. Testing includes data validation, performance testing, threshold calibration, and user-acceptance review. Implementation places the scoring logic and worklists into operational rhythm. Maintenance then monitors drift, threshold relevance, business adoption, and KPI movement.

This SDLC framing is important because churn prediction is not a one-time coding task. It is a living system. Customer behaviour changes, market conditions shift, and campaigns influence outcomes. Therefore, the bank should govern the churn model as a managed product with periodic review, not as a static academic exercise.

5. Analytical Techniques Applied

5.1 Descriptive Analytics

Descriptive analytics establishes the current-state picture. In the present case, the Power BI dashboard shows a baseline churn rate of 16.04%, churned customers of roughly 23K on the dashboard view, and strong month-to-month variation in churn behaviour. Although the record-level dataset used in the Excel file contains 3,165 churned customers out of 20,000 sample records, the dashboard is deliberately framed as a scaled banking view to mirror bank-level communication. This is analytically acceptable as long as the distinction is understood: the dashboard is a management layer and the Excel file is the modelling sample.

Monthly analysis from the dataset shows churn spikes in July, August, and January, with relatively lower levels in May and March. Such seasonality is useful because it allows the bank to time

campaigns and branch focus. Descriptive analytics also highlights portfolio composition. Individual customers dominate the portfolio and simultaneously exhibit the highest churn rate, which immediately makes them a priority segment for further diagnostic and predictive work.

5.2 Diagnostic Analytics

Diagnostic analytics seeks to explain why churn is happening. In this case, churn reasons recorded among churned customers are led by Stopped Using Account, High Fees, Better Offer from Another Bank, Poor Customer Service, App Issues, and Complaint. The largest category is voluntary disengagement—customers simply stopped using the account. That finding is strategically important because it implies the bank cannot treat churn only as a customer-service issue. A substantial portion of churn is behavioural disengagement, which means relationship relevance and digital usage need attention.

Table 2 shows the churn profile by customer type, while Table 3 summarises the leading recorded churn reasons.

Table 2. Churn profile by customer type

Customer_Type	Customers	Churned	Churn Rate %
Corporate	1571	47	2.99
Individual	14457	2791	19.31
Payroll	3972	327	8.23

Table 3. Leading recorded churn reasons among churned customers

Churn Reason	Customers	Share %
Stopped Using Account	1336	42.21
High Fees	783	24.74
Better Offer from Another Bank	359	11.34
Poor Customer Service	339	10.71
App Issues	223	7.05
Complaint	125	3.95

The diagnostic story becomes stronger when reasons are combined with behavioural features. Churned customers have lower average balances, much higher average risk scores, and longer gaps since last transaction. Low-balance customers show the highest churn rate by balance segment, while younger age groups also display comparatively higher churn rates. These patterns are consistent with a digital banking environment in which low-friction switching, weak primary-account attachment, and dissatisfaction with value proposition or convenience combine to increase attrition.

5.3 Predictive Analytics

Predictive analytics is the centrepiece of the case because it shifts the bank from describing churn to anticipating it. The logistic regression model uses both financial and behavioural variables, and on the simulated dataset it performs strongly. Table 4 summarises the evaluation metrics achieved on a holdout sample. The very high scores should be interpreted carefully because the dataset includes a risk score engineered to be highly informative. In real deployment, the bank should expect more moderate performance and should re-estimate model quality after excluding any features that may embed post-event information. Even with that caveat, the exercise remains valuable because it demonstrates how a predictive pipeline works and what management can do with it.

Table 4. Baseline logistic regression performance on the simulated dataset

Metric	Result
Accuracy	99.54%
Precision	98.61%
Recall	98.48%
F1-score	98.55%
ROC-AUC	0.9999

The practical meaning of these metrics is simple. Accuracy indicates the share of all predictions classified correctly. Precision indicates how many customers predicted to churn actually churned. Recall indicates how many of the true churners the model successfully identified. F1-score balances precision and recall, and ROC-AUC captures overall separation capability across thresholds. For business users, recall is especially important because missed churners represent lost intervention opportunities. Precision matters too because very low precision would waste retention resources on

customers who were never likely to leave. Threshold tuning therefore becomes a management question as much as a modelling question.

5.4 Prescriptive Analytics and Decision-Making Using Data Mining Techniques

Prescriptive analytics closes the loop by turning scores into decisions. In this case, the bank can define intervention thresholds based on both risk and value. For instance, a high-risk, high-CLV customer can be escalated to branch or relationship manager outreach within 24 hours. A high-risk, low-balance customer can receive automated digital nudges, fee relief, or service education prompts. A medium-risk payroll customer may require no price concession but may need service reassurance and app support.

This is where data mining techniques add value. Association of risk score, balance segment, customer type, and churn reason creates a richer decision layer than a raw risk score alone. A customer with high risk, low balance, and prior app issues should not receive the same treatment as a customer with high risk linked to pricing dissatisfaction. The bank should therefore treat the churn model as part of a broader next-best-action framework rather than a stand-alone probability engine.

5.5 KPIs, Metrics, and Equations

The case requires explicit KPI and metric treatment. The baseline formulas used in the report are presented below and are directly linked to management interpretation.

Churn Rate = (Number of Churned Customers ÷ Total Customers) × 100

Retention Rate = 100 – Churn Rate

Customer Lifetime Value (simplified operational form) = Average Annual Revenue per Customer × Expected Relationship Duration

Risk-Weighted Retention Priority = Churn Probability × CLV Index × Intervention Feasibility

Logistic Regression Probability:

$P(Churn=1) = 1 / (1 + e^{-(\beta 0 + \beta 1X1 + \beta 2X2 + ... + \beta nXn)})$

In the present study, X variables include age, balance, customer type, balance segment, branch, CLV, risk score, and days since last transaction. The bank can use these equations not only for reporting but for improving performance. For example, reducing average days since last transaction among medium-risk customers should lower modelled churn probability and therefore improve the retention KPI. Likewise, improving digital engagement among low-balance customers can affect both churn probability and campaign efficiency.

Table 5 sets out the main KPIs used in the case and the management action associated with each.

Table 5. Core churn-management KPIs and business use

KPI	Baseline value	Management use
Churn rate	15.82%	Track overall attrition pressure
Retention rate	84.18%	Monitor portfolio stability
Average CLV	EGP 338,106.96	Prioritise high-value saves
Average risk score	49.64	Identify urgency of intervention
Monthly churn trend	12-month series available	Time campaigns and branch staffing
Branch churn rate	Varies by branch	Escalate local issues and support
Customer-type churn rate	Varies by type	Differentiate retention strategy

5.6 Sample Dashboards, Charts, and Analytical Models

Figure 1 reproduces the supplied Power BI dashboard for Crédit Agricole Egypt. It serves as the baseline visual representation of the churn problem. Figure 2 adds a target-state monthly comparison using the same monthly series but assumes a 25% reduction in churned customers after AI-enabled intervention. This target-state is explicitly a scenario, not an already achieved outcome, and is used to quantify the managerial value of the proposed solution. Figure 3 provides a compact KPI view for the AI-enabled target state.

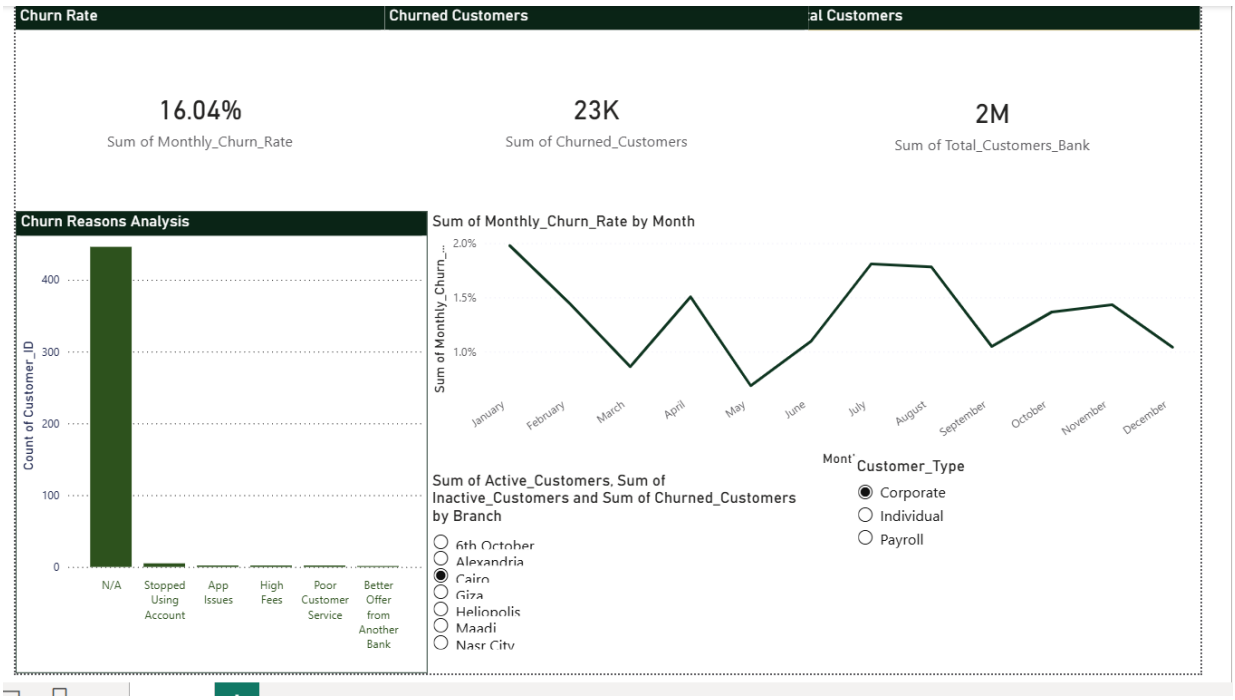

Figure 1. Current-state customer churn dashboard for Crédit Agricole Egypt (Power BI baseline view).

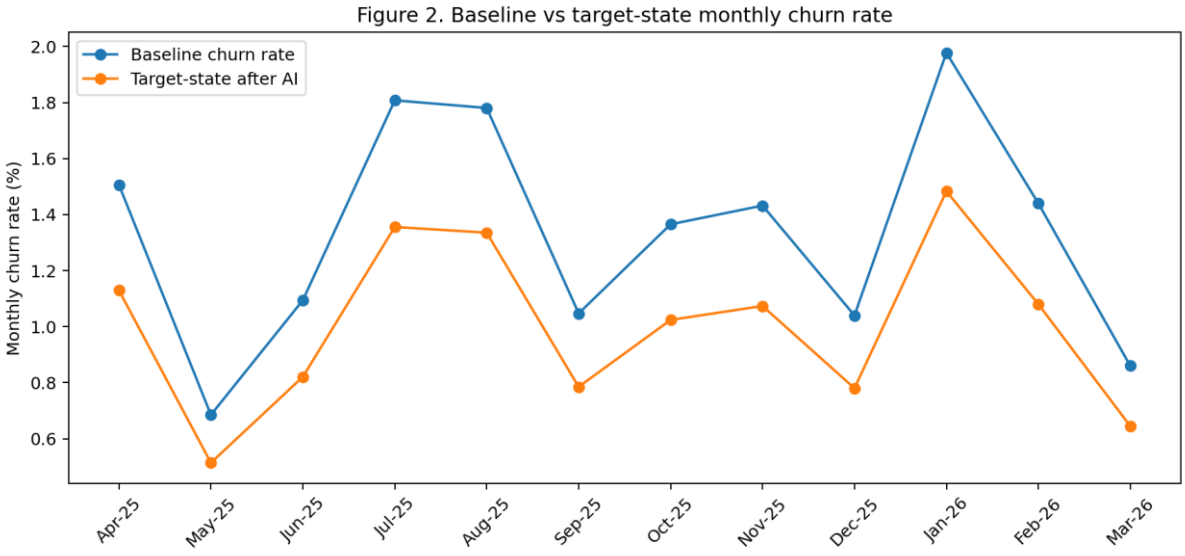

Figure 2. Baseline monthly churn rate versus target-state AI-enabled monthly churn rate assuming 25% fewer churned customers.

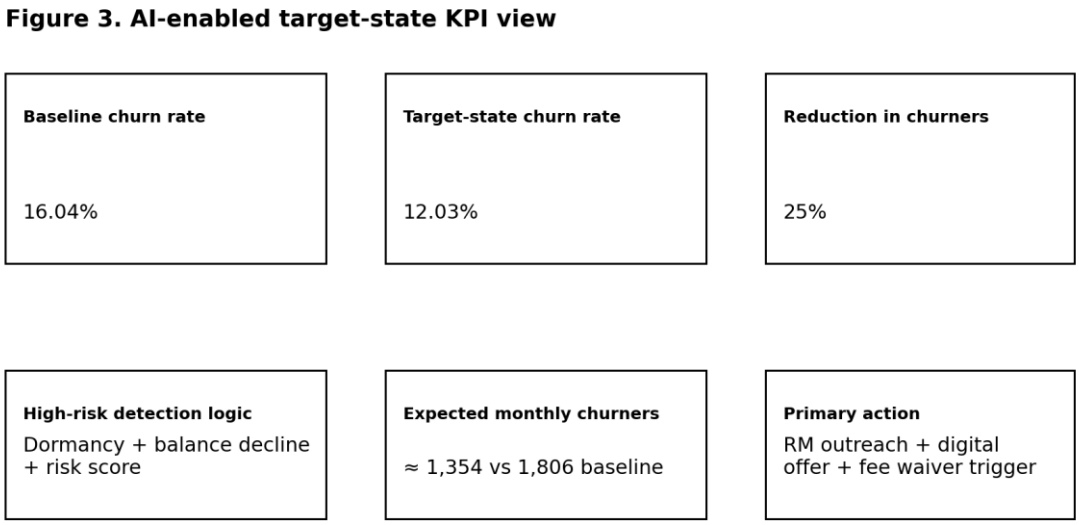

Figure 3. Illustrative AI-enabled target-state KPI view showing how intervention logic can change management focus from monitoring to action.

Table 6. Monthly baseline vs target-state churn scenario

Month	Baseline churned	Baseline churn rate	Target-state churned	Target-state churn rate
Apr-25	2211	1.51%	1658	1.13%
May-25	1001	0.69%	751	0.51%
Jun-25	1605	1.09%	1204	0.82%
Jul-25	2639	1.81%	1979	1.36%
Aug-25	2580	1.78%	1935	1.34%
Sep-25	1503	1.05%	1127	0.79%
Oct-25	1960	1.37%	1470	1.02%
Nov-25	2038	1.43%	1528	1.07%
Dec-25	1477	1.04%	1108	0.78%
Jan-26	2797	1.98%	2098	1.48%
Feb-26	2005	1.44%	1504	1.08%
Mar-26	1188	0.86%	891	0.65%

6. Digital Transformation Analysis

6.1 Changes in Business Processes and Workflows

The AI use case changes business processes in a fundamental way. In the pre-AI state, churn reporting is periodic and largely retrospective. Branches and support teams receive static summaries and decide case-by-case what to do, often after value leakage has already occurred. In the AI-enabled state, customer monitoring becomes continuous. New data from core banking and CRM refreshes customer status, the model recalculates churn risk, and the resulting score triggers branch or digital actions. This converts churn management from a reporting process into an intervention process.

From a workflow perspective, the change can be described as follows. Before AI, data extraction is manual, interpretation is human-only, and action is inconsistent. After AI, data extraction can be systematised, prioritisation is algorithmically supported, and action can be partly automated through RPA. Importantly, this does not remove human judgment. It upgrades the quality and timeliness of judgment.

6.2 Automation of Manual or Repetitive Tasks

Several repetitive tasks become candidates for automation once AI is integrated into the workflow. Customer lists can be generated automatically by risk band. Churn probability thresholds can populate branch task queues. Retention communications can be triggered automatically for pre-approved use cases. Escalation rules can be set so that high-value customers with rising risk receive human outreach, while low-value customers receive digital nudges. Monthly reporting can be replaced or supplemented with daily or weekly operational dashboards. The practical benefit is twofold: lower manual workload and greater consistency in response.

In the specific context of this case, automation also improves governance. Instead of relying on whether a relationship manager notices a concerning account pattern, the system guarantees that defined risk conditions generate a visible case. This reduces hidden churn risk and supports auditability.

6.3 Impact on Organizational Culture and Decision-Making

Digital transformation is not only about tools; it is about decision culture. AI pushes the bank toward evidence-based management. Branch and product teams become less reliant on intuition and

more reliant on measurable behaviour. That cultural change matters because churn can otherwise become a blame game—branches may blame product, product may blame service, and digital teams may blame customer inertia. Predictive analytics reframes the issue: where are the signals, where is the preventable risk, and which intervention has the best chance of success?

The cultural impact is also visible in accountability. Once branch- and segment-level risk is transparent, managers can no longer treat churn as a vague enterprise trend. They can see their local pattern and whether intervention improved outcomes. This strengthens performance management and encourages a more proactive service culture.

6.4 Customer Experience Improvements

From the customer's perspective, the target state should feel less like retention pressure and more like timely relevance. If the bank detects weak digital engagement and responds with app support, fee clarification, and more useful messaging, customer experience improves. If the bank detects rising churn risk among payroll customers and proactively resolves service friction, it can protect an embedded relationship before dissatisfaction grows. AI thus improves customer experience not by pushing more offers indiscriminately, but by helping the bank decide when, why, and how to intervene appropriately.

6.5 Comparison of Before vs After Digital Transformation

Table 7. Before-vs-after comparison of the churn-management model

Dimension	Before AI	After AI
Decision timing	After churn or late in disengagement	Before churn through risk scoring
Reporting style	Static, descriptive	Dynamic and predictive
Customer targeting	Generic segment campaigns	Customer-level prioritisation
Intervention execution	Manual and inconsistent	Automated plus human escalation
Data integration	Fragmented across systems	Integrated view across systems
Branch accountability	Weakly linked to risk	Explicit through risk dashboards
Customer experience	Reactive and less personalised	More timely and personalised

7. ERP Integration and Role

7.1 ERP Modules Involved and Organizational Relevance

Although the bank may not describe all systems under a classic manufacturing-style ERP vocabulary, the business role is equivalent. Core banking performs the transaction and account-keeping role, CRM performs the customer interaction and service role, and reporting layers provide management visibility. In the case narrative, Flexcube acts as the core banking backbone, while CRM holds customer identity and service-action context. Together they provide the minimum enterprise environment needed for AI-enabled churn management.

The relevant functional modules for this use case include account status management, transaction history, customer profile management, complaint handling, campaign management, and dashboard reporting. AI does not replace these modules. It links them by adding a probability layer that informs which cases deserve action first.

7.2 How AI Integrates with ERP Systems

AI integrates with ERP-like banking systems through a data pipeline rather than through a single screen. Historical and current data are extracted from the source systems, cleaned and structured in a common analytical model, scored through the churn algorithm, and then returned to operational tools where actions can be triggered. In practice, this means the CRM should display churn risk and recommended action, while branch dashboards should display local churn pressure and response performance.

This integration is essential because a predictive model has little business value if it sits outside the systems where teams actually work. Therefore, the design principle is not “build a model,” but “embed a model within an operating loop.”

7.3 Data Flow Between AI Tools and ERP

The end-to-end data flow can be formalised as a business process model:

Step 1: Core banking and CRM systems generate customer, balance, status, and behaviour records.
Step 2: Data is consolidated into an analytical layer or warehouse.
Step 3: Data quality checks and feature engineering create model-ready inputs.
Step 4: The AI model produces customer-level churn probabilities.

Step 5: Risk thresholds assign customers into action bands.

Step 6: CRM and branch systems receive the output with recommended interventions.

Step 7: Outcome data, such as response, reactivation, or continued inactivity, feeds back to the analytical layer for monitoring and retraining.

This flow is more than a technical diagram; it is the operational logic that allows AI to influence decisions.

7.4 Flow Chart, BPM, and Workflow Narrative

Flow Chart Narrative:

Core Banking / Flexcube + CRM + Digital Usage Data

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Data Integration Layer

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Data Cleansing and Feature Engineering

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Churn Prediction Model (AI)

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Risk Segmentation and Action Rules

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CRM / Branch Worklists / Automated Alerts

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Customer Outreach, Offer, Resolution, Monitoring

↓

Feedback Loop and Model Refresh

In BPM terms, the workflow starts with data capture, proceeds through analytical scoring, and ends with customer-facing intervention. The key design requirement is that each stage has an owner: IT and data teams own integration quality, analytics teams own feature and model quality, business teams own threshold approval and action policy, and branch or service teams own customer response. This explicit ownership reduces the common risk that AI projects become “everybody’s responsibility and therefore nobody’s responsibility.”

7.5 Benefits of ERP and AI Integration

The benefits of integrating AI with ERP-like banking systems are practical and measurable. First, the bank gains a single operating view of churn risk rather than disconnected lists. Second, interventions become faster because the model output appears where action occurs. Third, governance improves because actions can be monitored against model signals. Fourth, branch and central teams can coordinate around shared metrics instead of separate spreadsheets. Finally, model learning becomes stronger over time if response outcomes are captured and fed back into the system.

7.6 Challenges in ERP Implementation or Customization

The main implementation challenge is not algorithmic complexity but enterprise readiness. Source systems may use inconsistent customer identifiers, different update cycles, or incomplete fields. Branch workflows may not be designed to absorb risk-based task queues. Campaign systems may not support nuanced next-best-action logic. Customization is therefore often needed at the integration layer, in CRM views, and in reporting thresholds. Data governance becomes critical: poor data quality can make a technically strong model operationally weak.

This case therefore recommends phased implementation. The bank should begin with a minimum viable model and tightly defined workflow, then expand the solution only after adoption, quality, and business response are stabilised.

8. Business Impact and Value Creation

The business value of an AI-enabled churn programme must be assessed across financial, operational, and strategic dimensions. Financially, the most direct value comes from avoided churn. If the current dataset shows a baseline churn rate of 15.82%, then a 25% reduction in churned customers would move the rate closer to approximately 1.00% in the target-state scenario. Even without assigning a precise revenue-per-customer figure, that reduction means thousands of relationships preserved across a full-year horizon. In banking, that translates into protected balances, fee opportunities, and cross-sell potential.

Operationally, value arises from efficiency. Branch and CRM teams spend less time building lists and more time acting on prioritised cases. Management gains better visibility into where churn risk is concentrated, reducing the waste associated with undifferentiated campaigns. Strategically, AI

creates a capability advantage. A bank that can detect weakening relationships before attrition occurs is better positioned to defend market share, improve customer experience, and respond to competitors selectively rather than reactively.

The sustainability dimension is also important. An AI-enabled retention programme is scalable: once integrated into the data and CRM environment, it can be extended to additional products, channels, or relationship types. Thus the investment is not limited to one churn dashboard; it becomes part of the bank's broader digital operating model.

9. Risks, Ethical Issues, and Limitations

9.1 Data Privacy and Security Concerns

Customer churn analytics in banking inevitably deals with sensitive personal and financial information. The bank must therefore manage confidentiality, integrity, and availability as non-negotiable requirements. Data should be encrypted in transit and at rest; role-based access control should limit exposure of personally identifiable information; and audit trails should record who accessed which datasets and why. Where possible, development and testing should use masked data. These controls are not optional technical extras but part of the trust foundation of AI in banking.

9.2 Bias in AI Models

Algorithmic bias is a genuine risk. A model may inadvertently penalise groups that are overrepresented in certain churn patterns, such as younger or lower-balance customers, if business users interpret predictions as deterministic truths rather than risk signals. The bank must therefore separate prediction from unfair treatment. A high churn probability should trigger a retention review, not an exclusion or service downgrade. Fairness checks should compare error rates across key groups, and model governance should require business justification for any intervention policy that systematically affects one segment more than another.

9.3 Regulatory and Compliance Issues

Regulatory compliance extends beyond data security. The bank must be able to explain, at a reasonable level, how AI-informed decisions are being used. Because this case uses logistic regression as a baseline, explainability is comparatively strong. The bank can demonstrate which categories of features influence churn probability and how the model supports—not replaces—

business judgment. This is consistent with responsible AI principles that balance innovation with accountability.

9.4 Technology Adoption Risks and Limitations of AI and ERP Systems

Technology adoption risk is often underestimated. Staff may ignore scores, thresholds may be poorly calibrated, and campaigns may not be redesigned to use the insight well. ERP and CRM systems may not capture the feedback data needed for learning. There is also a material limitation in the present case: the dataset is realistic and analytically useful, but it is still simulated. This means the model metrics are more appropriate for demonstrating method than for claiming guaranteed production performance. For that reason, the bank should treat this case as a strong pilot blueprint, not as a final production validation.

9.5 ISO 27001 Clause Selection Relevant to the Topic

For this case, the most relevant ISO 27001 control areas include access control, information classification, cryptographic controls, logging and monitoring, supplier and system acquisition controls, and incident management. Table 8 maps selected control themes to the churn use case.

Table 8. Selected ISO 27001-aligned control areas relevant to the churn programme

ISO 27001 focus area	Relevance to churn use case
Access control	Restricts access to customer-level risk scores and PII
Information classification	Separates sensitive banking data from general reporting extracts
Cryptographic controls	Protects customer and transaction data in transit and storage
Logging and monitoring	Tracks data access, score generation, and intervention actions
Change management	Controls updates to model logic, thresholds, and workflows
Incident management	Supports response if customer data or model outputs are misused

10. Recommendations and Future Scope

The recommendations in this report follow directly from the data and the operating model proposed above. First, the bank should institutionalise a customer-level churn score refreshed on a defined schedule, ideally daily or weekly depending on system capability. Second, intervention thresholds should combine churn probability with customer value so that high-risk, high-value customers receive rapid human outreach. Third, the bank should redesign churn reason capture. The dashboard

currently shows that management-level reporting can still contain non-specific categories; over time, that weakens diagnostic power. Every closure, inactivity escalation, or major disengagement event should be coded with a structured reason wherever possible.

Fourth, digital engagement should become a central retention lever. Because behavioural disengagement is visible before closure, the bank should not wait for full churn to occur. It should trigger app support, service recovery, digital education, and convenience-based propositions when early indicators worsen. Fifth, branch-level dashboards should include not only outcome KPIs but action KPIs such as contact rate, conversion after outreach, and reactivation after intervention. Sixth, the bank should upgrade from a baseline logistic model to champion-challenger model management once production data becomes available; this may include random forests or gradient boosting if explainability and governance are maintained.

Future scope extends beyond the current case. The bank can integrate complaint text and call-center notes to bring unstructured data into the model through NLP. It can use unsupervised learning for behavioural segmentation and semi-supervised learning if labelled outcomes remain limited in some products. It can also extend the same framework to product-level churn, salary attrition, or deposit outflow prediction. Most importantly, it can build a feedback architecture in which every intervention outcome becomes new learning data, thereby improving the model and the workflow continuously.

Conclusion

This report has presented a complete AI-in-business case study on customer churn in digital banking, using Crédit Agricole Egypt as the applied context and realistic simulated banking data as the evidence base. The analysis shows that churn is measurable, diagnosable, and predictable. It is measurable through baseline KPIs and dashboard monitoring; diagnosable through behavioural, segment, and reason analysis; and predictable through supervised machine learning integrated into a bank workflow.

The central insight is that effective churn management requires more than a model. It requires joined-up data, an operational response process, ERP and CRM integration, customer-sensitive intervention logic, and governance aligned with privacy, fairness, and security. For Crédit Agricole

Egypt, the value of AI lies in enabling the bank to act earlier, target better, and learn faster. If deployed responsibly, the proposed approach can reduce attrition, protect customer lifetime value, improve branch prioritisation, and strengthen the bank's digital competitiveness. In that sense, the case is not simply about preventing customer loss; it is about using AI to redesign how the bank understands and protects customer relationships.

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